Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

PRESENTATION AND DISCLOSURE

Learning Outcome

- describe the presentation and disclosures relating to inventories and explain issues that analysts should consider when examining a company's inventory disclosures and other sources of information

The choice of inventory valuation method affects the financial statements. The financial statement items affected include cost of sales, gross profit, net income, inventories, current assets, and total assets. Therefore, the choice of inventory valuation method also affects financial ratios that contain these items. Ratios such as current ratio, return on assets, gross profit margin, and inventory turnover also are affected. As a consequence, analysts must carefully consider inventory valuation method differences when evaluating a company's performance over time or when comparing its performance with the performance of the industry or industry competitors. Additionally, the financial statement items and ratios may be affected by adjustments of inventory carrying amounts to net realizable value or current replacement cost.

Presentation and Disclosure

IFRS requires the following financial statement disclosures concerning inventory:

a. the accounting policies adopted in measuring inventories, including the cost formula (inventory valuation method) used;
b. the total carrying amount of inventories and the carrying amount in classifications (e.g., merchandise, raw materials, production supplies, work in progress, and finished goods) appropriate to the entity;
c. the carrying amount of inventories carried at fair value less costs to sell;
d. the amount of inventories recognized as an expense during the period (cost of sales);
e. the amount of any write-down of inventories recognized as an expense in the period;
f. the amount of any reversal of any write-down that is recognized as a reduction in cost of sales in the period;
g. the circumstances or events that led to the reversal of a write-down of inventories; and
h. the carrying amount of inventories pledged as security for liabilities.

Inventory-related disclosures under US GAAP are similar to these disclosures, except that requirements (f) and (g) are not relevant because US GAAP does not permit the reversal of prior-year inventory write-downs. US GAAP also requires the disclosure of significant estimates applicable to inventories and of any material amount of income resulting from the liquidation of LIFO inventory.

Inventory Ratios

Three ratios often used to evaluate the efficiency and effectiveness of inventory management are inventory turnover, days of inventory on hand, and gross profit margin.[5] These ratios are directly affected by a company's choice of inventory valuation method. Analysts should be aware, however, that many other ratios are also affected by the choice of inventory valuation method, although less directly. These include the current ratio, because inventory is a component of current assets; the return-on-assets ratio, because cost of sales is a key component in deriving net income and inventory is a component of total assets; and even the debt-to-equity ratio, because the cumulative measured net income from the inception of a business is an aggregate component of retained earnings.

The inventory turnover ratio measures the number of times during the year a company sells (i.e., turns over) its inventory. The higher the turnover ratio, the more times that inventory is sold during the year and the lower the relative investment of resources in

in the year, whereas days of inventory on hand is based on the ending inventory amount. In general, inventory turnover and the number of days of inventory on hand should be benchmarked against industry norms and compared across years.

A high inventory turnover ratio and a low number of days of inventory on hand might indicate highly effective inventory management. Alternatively, a high inventory ratio and a low number of days of inventory on hand could indicate that the company does not carry an adequate amount of inventory or that the company has written down inventory values. Inventory shortages could potentially result in lost sales or production problems in the case of the raw materials inventory of a manufacturer. To assess which explanation is more likely, analysts can compare the company's inventory turnover and sales growth rate with those of the industry and review financial statement disclosures. Slower growth combined with higher inventory turnover could indicate inadequate inventory levels. Write-downs of inventory could reflect poor inventory management. Minimal write-downs and sales growth rates at or above the industry's growth rates would support the interpretation that the higher turnover reflects greater efficiency in managing inventory.

A low inventory turnover ratio and a high number of days of inventory on hand relative to industry norms could be an indicator of slow-moving or obsolete inventory. Again, comparing the company's sales growth across years and with the industry and reviewing financial statement disclosures can provide additional insight.

The gross profit margin, the ratio of gross profit to sales, indicates the percentage of sales being contributed to net income as opposed to covering the cost of sales. Firms in highly competitive industries generally have lower gross profit margins than firms in industries with fewer competitors. A company's gross profit margin may be a function of its type of product. A company selling luxury products generally will have higher gross profit margins than a company selling staple products. The inventory turnover of the company selling luxury products, however, is likely to be much lower than the inventory turnover of the company selling staple products.

EXAMPLE 4

Single Company Illustration

Selected excerpts from the consolidated financial statements and notes to consolidated financial statements for Jollof Inc., a hypothetical telecommunications company providing networking and communications solutions. Exhibit 4 contains excerpts from the consolidated income statements, and Exhibit 5 contains excerpts from the consolidated balance sheets. Exhibit 6 contains excerpts from three of the notes to consolidated financial statements.

Note 1(a) discloses that Jollof's finished goods inventories and work in progress are valued at the lower of cost or net realizable value. Note 2(a) discloses that the impact of inventory and work in progress write-downs on Jollof's income before tax was a net reduction of EUR239 million in 2017, a net reduction of EUR156 million in 2016, and a net reduction of EUR65 million in 2015.[6] The inventory impairment loss amounts steadily increased from 2015 to 2017 and are included as a component, (additions)/reversals, of Jollof's change in valuation allowance as disclosed in Note 3(b) from Exhibit 6. Observe also that Jollof discloses its valuation allowance at 31 December 2017, 2016, and 2015 in Note 3(b) and details on the allocation of the allowance are included in Note 3(a). The EUR549 million valuation allowance is the total of a EUR528 million allowance for inventories and a EUR21 million allowance for work in progress on construction contracts. Finally, observe that the EUR1,845 million net value for inventories (excluding construction contracts) at 31 December 2017 in Note 3(a) reconciles with the balance sheet amount for inventories and work in progress, net, on 31 December 2017, as presented in Exhibit 5.

historical cost of the company's inventory by adding the inventory valuation allowance to the reported inventory carrying amount on the balance sheet. The valuation allowance increased in magnitude and as a percentage of gross inventory values from 2015 to 2017.

Exhibit 4:

Jollof Inc. Consolidated Income Statements (in millions of euros)

For years ended 31 December	2017	2016	2015
Revenues	14,267	14,945	10,317
Cost of sales	(9,400)	(10,150)	(6,900)
Gross profit	4,867	4,795	3,417
Administrative and selling expenses	(2,598)	(2,908)	(1,605)
Research and development costs	(2,316	(2,481)	(1,235)
Income from operating activities before restructuring costs, impairment of assets, gain/(loss) on disposal of consolidated entities, and post-retirement benefit plan amendments	(47)	(594)	577
Restructuring costs	(472)	(719)	(594)
Impairment of assets	(3,969)	(2,473)	(118)
Gain/(loss) on disposal of consolidated entities	(6)	—	13
Post-retirement benefit plan amendments	39	217	—
Income (loss) from operating activities	(4,455)	(3,569)	(122)
⋮	⋮	⋮	⋮
Income (loss) from continuing operations	(4,373)	(3,433)	(184)
Income (loss) from discontinued operations	28	512	133
Net income (loss)	(4,345)	(2,921)	51

Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

Exhibit 5:

Jollof Inc.Consolidated Balance Sheets (in millions of euros)

31 December	2017	2016	2015
⋮	⋮	⋮	⋮
Total non-current assets	10,703	16,913	21,559
Inventories and work in progress, net	1,845	1,877	1,898
Amounts due from customers on construction contracts	416	591	517
Trade receivables and related accounts, net	3,637	3,497	3,257
Advances and progress payments	83	92	73
⋮	⋮	⋮	⋮
Total current assets	12,238	11,504	13,629
Total assets	22,941	28,417	35,188
⋮	⋮	⋮	⋮
Retained earnings, fair value, and other reserves	(7,409)	(3,210)	(2,890)
⋮	⋮	⋮	⋮
Total shareholders' equity	4,388	9,830	13,711
Pensions, retirement indemnities, and other post-retirement benefits	4,038	3,735	4,577
Bonds and notes issued, long-term	3,302	3,794	4,117
Other long-term debt	56	40	123

Total non-current liabilities	8,736	9,471	11,219
Provisions	2,036	2,155	1,987
Current portion of long-term debt	921	406	975
Customers' deposits and advances	780	711	654
Amounts due to customers on construction contracts	158	342	229
Trade payables and related accounts	3,840	3,792	3,383
Liabilities related to disposal groups held for sale	—	—	1,349
Current income tax liabilities	155	59	55
Other current liabilities	1,926	1,651	1,625
Total current liabilities	9,817	9,117	10,257
Total liabilities and shareholders' equity	22,941	28,417	35,188

Exhibit 6:

Jollof Inc. Selected Notes to Consolidated Financial Statements

Note 1. Summary of Significant Accounting Policies

(a) Inventories and work in progress

Inventories and work in progress are valued at the lower of cost (including indirect production costs where applicable) or net realizable value. Net realizable value is the estimated sales revenue for a normal period of activity less expected completion and selling costs.

Note 2. Principal uncertainties regarding the use of estimates

(a) Valuation allowance for inventories and work in progress

Inventories and work in progress are measured at the lower of cost or net realizable value. Valuation allowances for inventories and work in progress are calculated based on an analysis of foreseeable changes in demand, technology, or the market, in order to determine obsolete or excess inventories and work in progress.

The valuation allowances are accounted for in cost of sales or in restructuring costs, depending on the nature of the amounts concerned.

	31 December		
(millions of euros)	2017	2016	2015
Valuation allowance for inventories and work in progress on construction contracts	(549)	(432)	318
Impact of inventory and work in progress write-downs on income (loss) before income tax related reduction of goodwill and discounted operations	(239)	(156)	(65)

Note 3. Inventories and work in progress

(a) Analysis of net value

(millions of euros)	2017	2016	2015
Raw materials and goods	545	474	455
Work in progress excluding construction contracts	816	805	632
Finished goods	1,011	995	1,109

Net value (excluding construction contracts)	1,845	1,877	1,898
Work in progress on construction contracts, gross*	184	228	291
Valuation allowance	(21)	(35)	(19)
Work in progress on construction contracts, net	163	193	272
Total, net	2,008	2,071	2,170

* Included in the amounts due from/to construction contracts.

(b) Change in valuation allowance

(millions of euros)	2017	2016	2015
At 1 January	(432)	(318)	(355)
(Additions)/reversals	(239)	(156)	(65)
Utilization	58	32	45
Changes in consolidation group			45
Net effect of exchange rate changes and other changes	63	10	12
At 31 December	(549)	(432)	(318)

Rounding differences may result in totals that are slightly different from the sum and from corresponding numbers in the note.

Rate Your Confidence

Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

1. Calculate Jollof's inventory turnover, number of days of inventory on hand, gross profit margin, current ratio, debt-to-equity ratio, and return on total assets for 2017 and 2016 based on the numbers reported. Use an average for inventory and total asset amounts and year-end numbers for other ratio items. For debt, include only bonds and notes issued, long-term; other long-term debt; and current portion of long-term debt.

Hide Solution

Solution:

The financial ratios are as follows:

	2017	2016
Inventory turnover ratio	5.05	5.38
Number of days of inventory on hand	72.3 days	67.8 days
Gross profit margin	34.1%	32.1%
Current ratio	1.25	1.26
Debt-to-equity ratio	0.98	0.43
Return on total assets	–16.9%	–9.2%

Inventory turnover ratio = Cost of sales ÷ Average inventory

2017 inventory turnover ratio = 5.05 = 9,400 ÷ [(1,845 + 1,877) ÷ 2]

2016 inventory turnover ratio = 5.38 = 10,150 ÷ [(1,877 + 1,898) ÷ 2]

Number of days of inventory = 365 days ÷ Inventory turnover ratio

2017 number of days of inventory = 72.3 days = 365 days ÷ 5.05

2016 number of days of inventory = 67.8 days = 365 days ÷ 5.38

Gross profit margin = Gross profit ÷ Total revenue

2017 gross profit margin = 34.1% = 4,867 ÷ 14,267

2016 gross profit margin = 32.1% = 4,795 ÷ 14,945

Current ratio = Current assets ÷ Current liabilities

2017 current ratio = 1.25 = 12,238 ÷ 9,817

2016 current ratio = 1.26 = 11,504 ÷ 9,117

Rate Your Confidence

Return on assets = Net income ÷ Average total assets

2017 return on assets = –16.9% = –4,345 ÷ [(22,941 + 28,417) ÷ 2]

2016 return on assets = –9.2% = –2,921 ÷ [(28,417 + 35,188) ÷ 2]

2. Based on the answer to question 1, comment on the changes from 2016 to 2017.

Hide Solution

Solution:

From 2016 to 2017, the inventory turnover ratio declined and the number of days of inventory increased by 4.5 days. Jollof appears to be managing inventory less efficiently. The gross profit margin improved by 2.0 percent, from 32.1 percent in 2016 to 34.1 percent in 2017. The current ratio is relatively unchanged from 2016 to 2017. The debt-to-equity ratio has risen significantly in 2017 compared to 2016. Although Jollof's total debt has been relatively stable during this time period, the company's equity has been declining rapidly because of the cumulative effect of its net losses on retained earnings.

The return on assets is negative and deteriorated in 2017 compared to 2016. A larger net loss and lower total assets in 2017 resulted in a higher negative return on assets. The analyst should investigate the underlying reasons for the sharp decline in Jollof's return on assets. From Exhibit 4, it is apparent that Jollof's gross profit margins were insufficient to cover the administrative and selling expenses and research and development costs in 2016 and 2017. Large restructuring costs and asset impairment losses contributed to the loss from operating activities in both 2016 and 2017.

Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

3. If Jollof had used the weighted average cost method instead of the FIFO method during 2017, 2016, and 2015, what would be the effect on Jollof's reported cost of sales and inventory carrying amounts? What would be the directional impact on the financial ratios that were calculated for Jollof in Question 1?

Hide Solution

Solution:

If inventory replacement costs were increasing during 2015, 2016, and 2017 (and inventory quantity levels were stable or increasing), Jollof's cost of sales would have been higher and its gross profit margin would have been lower under the weighted average cost inventory method than what was reported under the FIFO method (assuming no inventory write-downs that otherwise would neutralize the differences between the inventory valuation methods). FIFO allocates the oldest inventory costs to cost of sales; the reported cost of sales would be lower under FIFO given increasing inventory costs. Inventory carrying amounts would be higher under the FIFO method than under the weighted average cost method because the more recently purchased inventory items would be included in inventory at their higher costs (again assuming no inventory write-downs that otherwise would neutralize the differences between the inventory valuation methods). Consequently, Jollof's reported gross profit, net income, and retained earnings would also be higher for those years under the FIFO method.

The effects on ratios are as follows:

- The inventory turnover ratios would all be higher under the weighted average cost method because the numerator (cost of sales) would be

average cost method because the inventory turnover ratios would be higher.

- The gross profit margin ratios would all be lower under the weighted average cost method because cost of sales would be higher under the weighted average cost method than under the FIFO method.
- The current ratios would all be lower under the weighted average cost method because inventory carrying values would be lower than under the FIFO method (current liabilities would be the same under both methods).
- The return-on-assets ratios would all be lower under the weighted average cost method because the incremental profit added to the numerator (net income) has a greater impact than the incremental increase to the denominator (total assets). By way of example, assume that a company has EUR3 million in net income and EUR100 million in total assets using the weighted average cost method. If the company reports another EUR1 million in net income by using FIFO instead of weighted average cost, it would then also report an additional EUR1 million in total assets (after tax). Based on this example, the return on assets is 3.00 percent (EUR3/EUR100) under the weighted average cost method and 3.96 percent (EUR4/EUR101) under the FIFO method.
- The debt-to-equity ratios would all be higher under the weighted average cost method because retained earnings would be lower than under the FIFO method (again assuming no inventory write-downs that otherwise would neutralize the differences between the inventory valuation methods).
- Conversely, if inventory replacement costs were decreasing during 2015, 2016, and 2017 (and inventory quantity levels were stable or increasing), Jollof's cost of sales would have been lower and its gross profit and inventory would have been higher under the weighted average cost method than were reported under the FIFO method (assuming no inventory write-downs that otherwise would neutralize the differences between the inventory valuation methods). As a result, the ratio assessment that was performed above would result in directly opposite conclusions.

Rate Your Confidence

Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

{FLDS} FIELD SEARCH

The Bloomberg FLDS function allows you to view Bloomberg's data fields, including inventory statistics.

> "FLDS <GO> is your hub for exploring Bloomberg data, providing a breadth of fields for legacy APIs, calcrts, and the Bloomberg Query Language (BQL) in one centralized location." -Bloomberg

To explore this function, type FLDS into the Bloomberg command line and press Enter or <GO>. Type a security—such as Industria de Diseno Textil SA (or ITX SM Equity)—into the top left amber field, and press Enter or <GO>. Then type "inventory" into the Enter

Search for Fields | Selected Fields (0)

inventory | View Ranked | Filter Equity | Field Type All

	ID	Mnemonic	Ovrd	Value
1)	BS013	BS_INVENTORIES		3,191.00
2)	RR056	INVENT_TURN		3.98
3)	RR160	INVENT_DAYS		91.66
4)	BS137	INVTRY_FINISHED_GOODS		2,898.00
5)	BS135	INVTRY_RAW_MATERIALS		228.00
6)	BS136	INVTRY_IN_PROGRESS		65.00
7)	DX828	INVENTORY_METHOD		
8)	CF109	CF_CHANGE_IN_INVENTORIES		-193.00
9)	RR892	INVENTORY_GROWTH		4.63
10)	A0048	ARD_INVENTORY		3,191.00
11)	A0136	ARD_CHANGE_IN_INVENTOR...		-193.00
12)	RX014	INVENTORY_GROWTH_TO_S...		0.28
13)	RR731	INVENT_TO_SALES		11.01
14)	BS147	BS_OTHER_INV		0.00
15)	F0473	INVENTORY_5_YEAR_GROWTH		3.51
16)	F0495	INVENTORY_SEQUENTIAL_G...		4.90
17)	RR258	INV_TO_CASH_DAYS		87.35

*Fields that appear in white provide streaming data.

© 2023 Bloomberg L.P.

Please provide feedback on this Bloomberg feature.

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Discuss | Filter

Discussion

What do you want to discuss... (required)

ZE

In the practice question i believe 5 , where we had to calculate the inventory turnover ratios for the three cases (we had to determine which method a person used kinda), i noticed they use as denominator the inventory of the year itself and not the average of year one and year 2, which formula is the "More Correct One " ??

Created 5 months ago by Zakaria El Yassini — 0 replies | Last Activity: 5 months ago

Reply to this Comment

MO

In practice questions for the ZP Corporation. Q7: In Exhibit 2, the Industry and Business Risk excerpt states that, “Increased competition may lead to lower unit sales and excess production capacity and excess inventory. This may result in a further downward price pressure.” The downward price pressure could lead to inventory that is valued above current market prices or net realizable value. Any write-downs of inventory are least likely to have a significant effect on the inventory valued using:

A. weighted average cost.
B. first-in, first-out (FIFO).
C. last-in, first-out (LIFO).

The systems says that the correct answer is B, whereas when it gives the answer it says that LIFO is the correct answer. I'm confused.

"If prices have been decreasing, write-downs under FIFO are least likely to have a significant effect because the inventory is valued at closer to the new, lower prices. Typically, inventories valued using LIFO are less likely to incur inventory write-downs than inventories valued using weighted average cost or FIFO. Under LIFO, the oldest costs are reflected in the inventory carrying value on the balance sheet. Given increasing inventory costs, the inventory carrying values under the LIFO method are already conservatively presented at the oldest and lowest costs. Thus, it is far less likely that inventory write-downs will occur under LIFO; and if a write-down does occur, it is likely to be of a lesser magnitude."

Created 6 months ago by Maxim Ovchinnikov — 1 reply | Last Activity: 5 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic